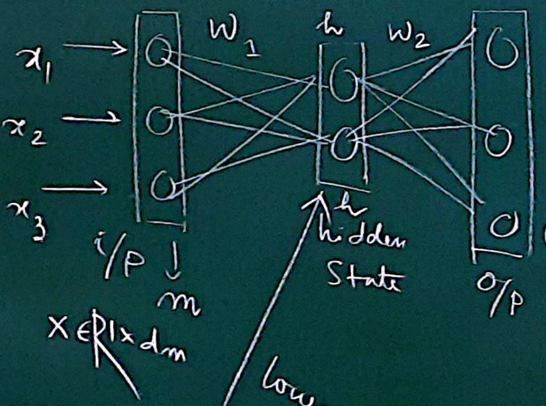
Autoencoder

↓

Dimensionality Reduction (PCA)

W_1 $d_m \times d_h$ $W_2 \in R^{d_h \times d_m}$

$h < m$



$a(W_1^T x + b_1)\big|_{1 \times d_h}$

$O = a\left(W_2^T\left(a(W_1^T x + b_1) + b_2\right)\right)\Big|_{1 \times d_m}$

$X \in R^{1 \times d_m}$

low-dim representation of x

under-complete AE

- Dimension of I/P layer and O/P layer should be the same

\# of neurons

- Self-supervised model

(ground-truth data not needed)

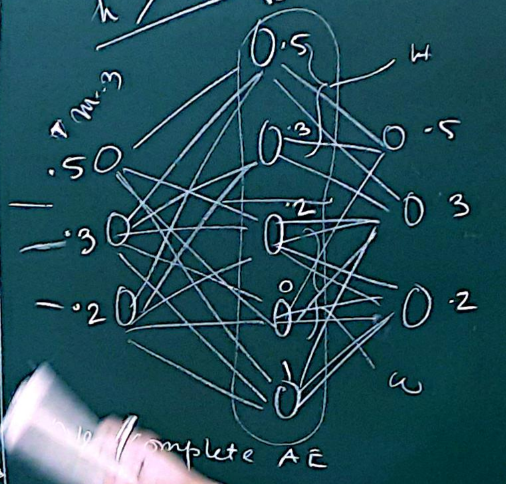
$loss(O, X) = CE(O, X)$

BMI

$f(H\,W)$

$h > m$

Auto encoder



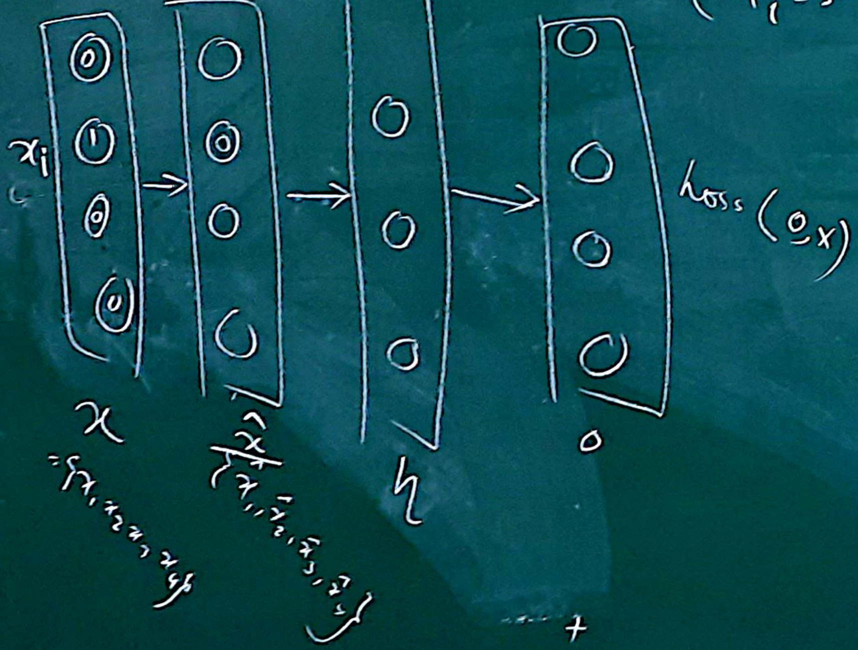
Autoencoder

↓

Denoising Autoencoder

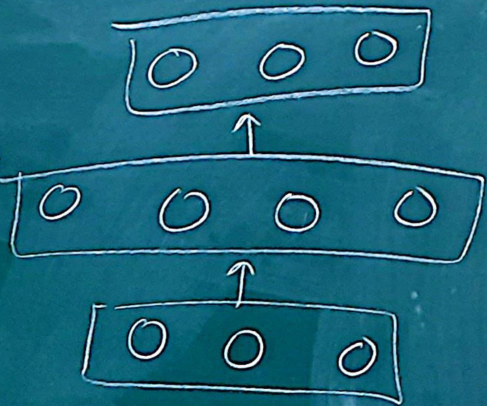
$$p(\hat{x}_i = x_i \mid x_i) = q$$

$$p(\hat{x}_i = 1 - x_i \mid x_i) = 1 - q$$



Sparse Autoencoder

Overfitting



Autoencoder

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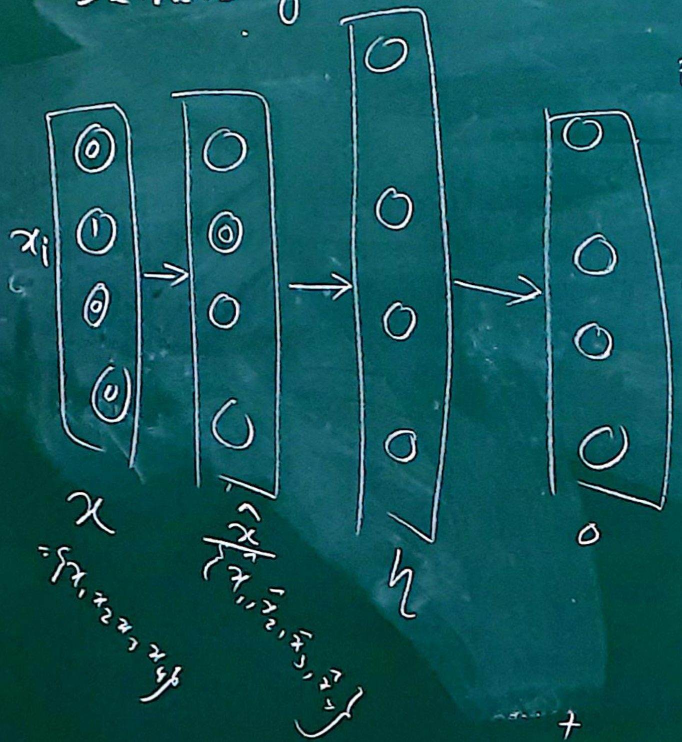
Denoising Autoencoder



loss (o, x)

$p(\hat{x}_i = x_i | x_i) = q$

$p(\hat{x}_i = 1 - x_i | x_i) = 1 - q$

loss = CE( ) +

$l = CE(x, \hat{x}) + Reg$

Sparse Autoencoder

Overfitting

higher parameter

0.005

sparsity

Reg:

$$\sum\left(\rho\log\frac{\rho}{\rho_i} + (1-\rho)\log\frac{1-\rho}{1-\rho_i}\right)$$

$$+\sum \rho\log\rho - \rho\log\rho_i + \log\frac{(1-\rho)}{1-\rho_i} - \rho\log\frac{1-\rho}{1-\rho_i}$$

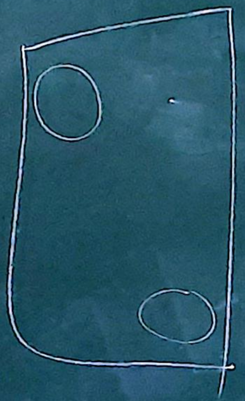
$$+\log(1-\rho) - \log(1-\rho_i)$$

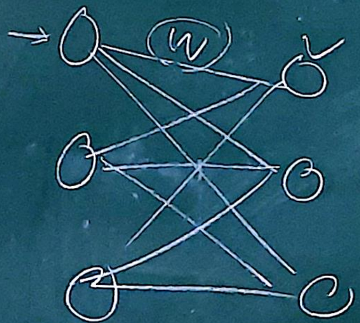
$$-\rho\log(1-\rho)$$

$$+\rho\log(1-\rho_i)$$

- The movie is bad. It is not good.

- The movie is good. It is not bad





1. Sparsity
2. Sharing of infor

Convolution

Commutative

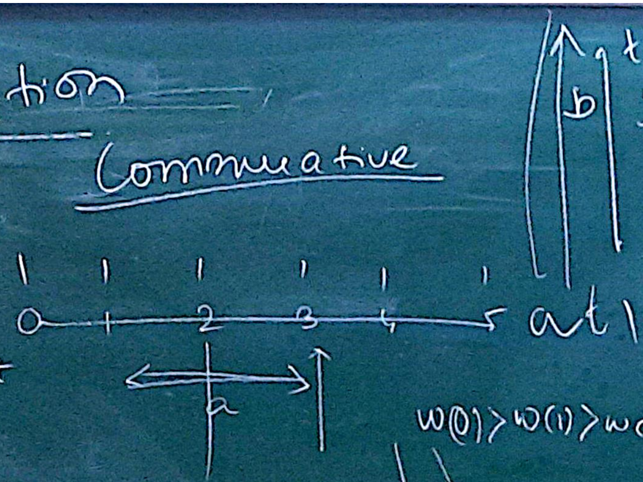
$x(t)$ = position of the vehical at t

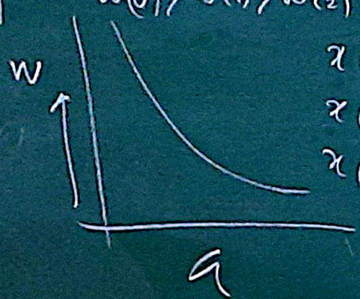
Sensor

a

$w(\ )$

$\int x(t - a \ldots$





$w(0) > w(1) > w(2) \ldots \quad a = 0$

$x(t)\, w(0)$

$x(t-1)\, w(1)$

$x(t-2)\, w(2)$

Smoothed estimation of the position using $x(\ )$ and $w(\ )$

$$s(t) = \int_{a}^{a} x(t-a)\, w(a)\, da$$

$$= \sum_{a=1}^{\infty} x(t-a)\, w(a)$$

$$= S(t) = (x * w)(t)$$

$$= (w * x)(t)$$

Convolution $\sum_{b=1}^{\infty}\sum_{a=1}^{\infty} x(t_1 - a, t_2 - b) \times w(a, b)$

1	0	1
1	0	1
0	0	1

$W1_{3\times 3}$

Kernel / filter

I

$\sigma(w^T x + b)$

0.5	0.2	.3	0.2	0.8
0.3	0.2	0.1	.05	.06
.08	0.7	0.1	0.2	0.3
0.4	0.7	0.5	0.1	

Sum

t2

0.5	0	0.3
0.3	0	0.1
0.	0	0.1